

§4.2 Richardson’s Extrapolation

General principle.

If  $N(h) = M + K_1h + K_2h^2 + \cdots$  approximates  $M$  with error  $O(h)$ , then combining  $N(h)$  and  $N(h/2)$  to cancel the leading  $K_1h$  gives  $O(h^2)$ :

$$N_2(h) = N(h/2) + [N(h/2) - N(h)].$$

Even-power expansion (e.g. centered differences, comp. Trap.).

If  $N(h) = M + K_1h^2 + K_2h^4 + \cdots$ :

$$N_j(h) = N_{j-1}(h/2) + \frac{N_{j-1}(h/2) - N_{j-1}(h)}{4^{j-1} - 1}, \quad j \geq 2.$$

Each step kills the next  $h^{2(j-1)}$  term:  $N_j(h) - M = O(h^{2j})$ .

Romberg integration.

Apply to composite Trapezoidal  $T_n$ :

$$R_{k,1} = T_{2^{k-1}}, \quad R_{k,j} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}.$$

$R_{k,2}$  is Simpson’s,  $R_{k,3}$  is Boole’s. Triangular table; cost  $\approx 2^{k-1} + 1$  function values.

Three-pt midpoint Richardson example.

$$\phi(h) = \frac{f(x_0+h) - f(x_0-h)}{2h} = f'(x_0) + \sum K_i h^{2i} \Rightarrow f'(x_0) \approx \frac{4\phi(h/2) - \phi(h)}{3}, \text{ error } O(h^4).$$

§4.3 Numerical Integration

Definition 4.1 (degree of precision).

Largest integer  $n$  with  $\int_a^b x^k dx = \sum_i a_i x_i^k$  for  $k = 0, \dots, n$ . Equivalently  $E(x^k) = 0$  for  $k \leq n$  and  $E(x^{n+1}) \neq 0$ . By linearity of  $E$ : deg. precision  $\geq n \iff$  formula exact for all  $\mathcal{P}_n$ .

Trapezoidal rule

( $n = 1$ ,  $h = b - a$ , deg. prec. 1):

$$\int_a^b f dx = \frac{h}{2} [f(a) + f(b)] - \frac{h^3}{12} f''(\xi), \quad \xi \in (a, b).$$

Simpson’s rule

( $n = 2$ ,  $h = \frac{b-a}{2}$ ,  $x_1 = \frac{a+b}{2}$ , deg. prec. 3):

$$\int_a^b f dx = \frac{h}{3} [f(a) + 4f(x_1) + f(b)] - \frac{h^5}{90} f^{(4)}(\xi).$$

Simpson’s 3/8 rule

( $n = 3$ ,  $h = \frac{b-a}{3}$ , deg. prec. 3):

$$\int_a^b f dx = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] - \frac{3h^5}{80} f^{(4)}(\xi).$$

Composite Trapezoidal

( $h = (b - a)/n$ ,  $x_j = a + jh$ ):

$$\int_a^b f = \frac{h}{2} \left[ f(a) + 2 \sum_{j=1}^{n-1} f(x_j) + f(b) \right] - \frac{(b-a)h^2}{12} f''(\mu).$$

Composite Simpson’s

( $n$  even,  $h = (b - a)/n$ ):

$$\int_a^b f = \frac{h}{3} \left[ f(a) + 2 \sum_{\text{even}} f(x_j) + 4 \sum_{\text{odd}} f(x_j) + f(b) \right] - \frac{(b-a)h^4}{180} f^{(4)}(\mu).$$

Composite Midpoint

( $n$  even,  $h = (b - a)/(n + 2)$ ,  $x_j = a + (j + 1)h$ ):

$$\int_a^b f = 2h \sum_{j=0}^{n/2} f(x_{2j+1}) + \frac{(b-a)h^2}{6} f''(\mu).$$

Stability.

All these are *stable*: round-off in nodes amplifies by  $\leq \sum |a_i| = b - a$  (no cancellation).

Method of undetermined coefficients.

For  $\int_a^b f dx \approx \sum a_i f(x_i)$  with prescribed nodes: impose exactness on  $1, x, x^2, \dots$  to get linear system for  $a_i$ ; one extra monomial determines the error constant  $k f^{(p)}(\xi)$ .

§4.7 Gaussian Quadrature

Idea.

Choose both nodes  $x_i$  and weights  $c_i$  ( $n$  each  $\Rightarrow 2n$  free parameters)  $\Rightarrow$  deg. precision  $2n - 1$  (max possible).

Theorem 4.7 (nodes & weights on  $[-1, 1]$ ).

Nodes  $x_1, \dots, x_n$  are roots of  $n$ th Legendre polynomial  $P_n$ ; weights

$$c_i = \int_{-1}^1 \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx > 0.$$

Legendre polynomials

(orthogonal on  $[-1, 1]$ , monic):  $P_0 = 1$ ,  $P_1 = x$ ,  $P_2 = x^2 - \frac{1}{3}$ ,  $P_3 = x^3 - \frac{3}{5}x$ ,  $P_4 = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$ .

Table 4.12 (nodes  $r_{n,i}$ , weights  $c_{n,i}$ ).

| $n$ | roots $\pm r_{n,i}$                                      | coefficients $c_{n,i}$                       |
|-----|----------------------------------------------------------|----------------------------------------------|
| 2   | $\pm 0.5773502692$                                       | 1.0000000000                                 |
| 3   | $\pm 0.7745966692$<br>0.0000000000                       | 0.5555555556<br>0.8888888889                 |
| 4   | $\pm 0.8611363116$<br>$\pm 0.3399810436$                 | 0.3478548451<br>0.6521451549                 |
| 5   | $\pm 0.9061798459$<br>$\pm 0.5384693101$<br>0.0000000000 | 0.2369268850<br>0.4786286705<br>0.5688888889 |

Arbitrary  $[a, b]$ : change of variable.

$$x = \frac{(b-a)t + (a+b)}{2}, \quad dx = \frac{b-a}{2} dt:$$

$$\int_a^b f(x) dx = \frac{b-a}{2} \int_{-1}^1 f\left(\frac{(b-a)t + (a+b)}{2}\right) dt \approx \frac{b-a}{2} \sum_{i=1}^n c_i f\left(\frac{(b-a)x_i + (a+b)}{2}\right).$$

Error.

Exact for polys of degree  $\leq 2n - 1$ . For  $f \in C^{2n}[a, b]$ :

$$E_n[f] = \frac{(b-a)^{2n+1} (n!)^4}{(2n+1)! [(2n)!]^3} f^{(2n)}(\xi), \quad \xi \in (a, b).$$

Generally far more accurate than Newton-Cotes for the same number of evaluations.

## §5.1 Theory of Initial-Value Problems

### IVP.

$y'(t) = f(t, y)$ ,  $a \leq t \leq b$ ,  $y(a) = \alpha$ .

**Definition (Lipschitz in  $y$ ).**

$f$  satisfies a Lipschitz condition on  $D \subset \mathbb{R}^2$  in  $y$  if  $\exists L > 0$ :

$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2|, \quad \forall (t, y_1), (t, y_2) \in D.$$

**Theorem 5.3 (sufficient).**

If  $D \subset \mathbb{R}^2$  is convex and  $|\partial f / \partial y| \leq L$  on  $D$ , then  $f$  is Lipschitz in  $y$  on  $D$  with constant  $L$ . (Sufficient, not necessary.)

**Theorem 5.4 (Existence & uniqueness).**

If  $D = \{(t, y) : a \leq t \leq b, y \in \mathbb{R}\}$ ,  $f$  is continuous on  $D$ , and  $f$  is Lipschitz in  $y$  on  $D$ , then the IVP has a unique solution  $y(t)$  on  $[a, b]$ .

**Definition (well-posed).**

The IVP is well-posed if (i) it has a unique solution; (ii)  $\forall \varepsilon > 0, \exists k(\varepsilon) > 0$  s.t. for  $|\varepsilon_0| < \varepsilon$  and any  $\delta(t)$  continuous with  $|\delta(t)| < \varepsilon$  on  $[a, b]$ , the perturbed problem  $z' = f(t, z) + \delta(t)$ ,  $z(a) = \alpha + \varepsilon_0$  has a unique  $z(t)$  with  $|z(t) - y(t)| < k(\varepsilon)\varepsilon$ .

**Theorem 5.6.**

If  $f$  is continuous and Lipschitz in  $y$  on  $D = \{a \leq t \leq b, y \in \mathbb{R}\} \Rightarrow$  IVP is well-posed.

## §5.2 Euler's Method

**Mesh and step.**

$$t_i = a + ih, \quad i = 0, \dots, N, \quad h = (b - a)/N.$$

**Formula**

(Taylor order 1):

$$w_0 = \alpha, \quad w_{i+1} = w_i + h f(t_i, w_i).$$

Local truncation error  $\tau_{i+1}(h) = \frac{h}{2} y''(\xi_i) = O(h)$ .

**Theorem 5.9 (error bound).**

If  $f$  Lipschitz with constant  $L$  on  $D = \{a \leq t \leq b, y \in \mathbb{R}\}$  and  $|y''(t)| \leq M$  on  $[a, b]$ , then for the exact solution  $y$ :

$$|y(t_i) - w_i| \leq \frac{hM}{2L} \left[ e^{L(t_i - a)} - 1 \right], \quad i = 0, \dots, N.$$

**Theorem 5.10 (with round-off).**

If  $|\delta_i| \leq \delta$  then

$$|y(t_i) - u_i| \leq \frac{1}{L} \left( \frac{hM}{2} + \frac{\delta}{h} \right) \left[ e^{L(t_i - a)} - 1 \right] + |\delta_0| e^{L(t_i - a)}.$$

Optimal step:  $h^* = \sqrt{2\delta/M}$  (minimizes  $hM/2 + \delta/h$ ).

## §5.3 Higher-Order Taylor Methods

**Order  $n$  method.**

$$w_0 = \alpha, \quad w_{i+1} = w_i + h T^{(n)}(t_i, w_i),$$

$$T^{(n)}(t, w) = f(t, w) + \frac{h}{2} f'(t, w) + \dots + \frac{h^{n-1}}{n!} f^{(n-1)}(t, w),$$

where  $f^{(k-1)}(t, y(t)) = \frac{d^k y}{dt^k}$  along the solution curve. Local truncation error  $\tau_{i+1}(h) = \frac{h^n}{(n+1)!} y^{(n+1)}(\xi) = O(h^n)$ .

**Computing  $f', f'',$  etc.**

Differentiate  $y' = f(t, y)$  via chain rule:

$$f' = f_t + f_y \cdot f, \quad f'' = f_{tt} + 2f_{ty}f + f_{yy}f^2 + f_y(f_t + f_y f).$$

**Drawback.**

Requires symbolic derivatives of  $f$  — Runge-Kutta replaces these with extra  $f$ -evaluations.

## §5.4 Runge-Kutta Methods

**Midpoint method (RK2)**

LTE  $O(h^2)$ :

$$w_{i+1} = w_i + h f\left(t_i + \frac{h}{2}, w_i + \frac{h}{2} f(t_i, w_i)\right).$$

**Modified Euler / Heun-trapezoid (RK2)**

LTE  $O(h^2)$ :

$$w_{i+1} = w_i + \frac{h}{2} \left[ f(t_i, w_i) + f\left(t_{i+1}, w_i + h f(t_i, w_i)\right) \right].$$

**Heun's method (RK3)**

LTE  $O(h^3)$ :

$$w_{i+1} = w_i + \frac{h}{4} \left[ f(t_i, w_i) + 3f\left(t_i + \frac{2h}{3}, w_i + \frac{2h}{3} f(t_i + \frac{h}{3}, w_i + \frac{h}{3} f(t_i, w_i))\right) \right].$$

**Runge-Kutta order 4 (RK4)**

LTE  $O(h^4)$  (most-used):

$$\begin{aligned} k_1 &= h f(t_i, w_i), \\ k_2 &= h f\left(t_i + \frac{h}{2}, w_i + \frac{k_1}{2}\right), \\ k_3 &= h f\left(t_i + \frac{h}{2}, w_i + \frac{k_2}{2}\right), \\ k_4 &= h f(t_{i+1}, w_i + k_3), \\ w_{i+1} &= w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

**Cost vs. order (Butcher).**

| evals/step $n$ | 2        | 3        | 4        | $\geq 5$          |
|----------------|----------|----------|----------|-------------------|
| best LTE       | $O(h^2)$ | $O(h^3)$ | $O(h^4)$ | $\leq O(h^{n-1})$ |

RK4 dominates RK2 with  $h$  vs.  $h/2$  and Euler with  $h$  vs.  $h/4$  for the same total  $f$ -cost.

**Global error.**

If LTE =  $O(h^p)$  and  $f$  is Lipschitz, global error  $|y(t_i) - w_i| = O(h^p)$  as well (one order is lost from local  $\rightarrow$  global only if you confuse the two — for one-step methods of order  $p$ , both LTE and global err are  $O(h^p)$ ).

### Quick reference: error orders

| Method                  | Error                       | Cost/step     |
|-------------------------|-----------------------------|---------------|
| Trapezoidal (single)    | $O(h^3)$                    | 2 evals       |
| Simpson's (single)      | $O(h^5)$                    | 3 evals       |
| Comp. Trapezoidal       | $O(h^2)$                    | $n + 1$       |
| Comp. Simpson           | $O(h^4)$                    | $n + 1$       |
| Comp. Midpoint          | $O(h^2)$                    | $n/2 + 1$     |
| Romberg ( $k$ levels)   | $O(h^{2k})$                 | $2^{k-1} + 1$ |
| Gauss-Legendre $n$ -pt  | $O(h^{2n})$ , deg. $2n - 1$ | $n$           |
| Euler                   | $O(h)$                      | 1             |
| Taylor order $n$        | $O(h^n)$                    | 1 (+ derivs)  |
| RK2 (midpt./mod. Euler) | $O(h^2)$                    | 2             |
| RK3 (Heun)              | $O(h^3)$                    | 3             |
| RK4                     | $O(h^4)$                    | 4             |